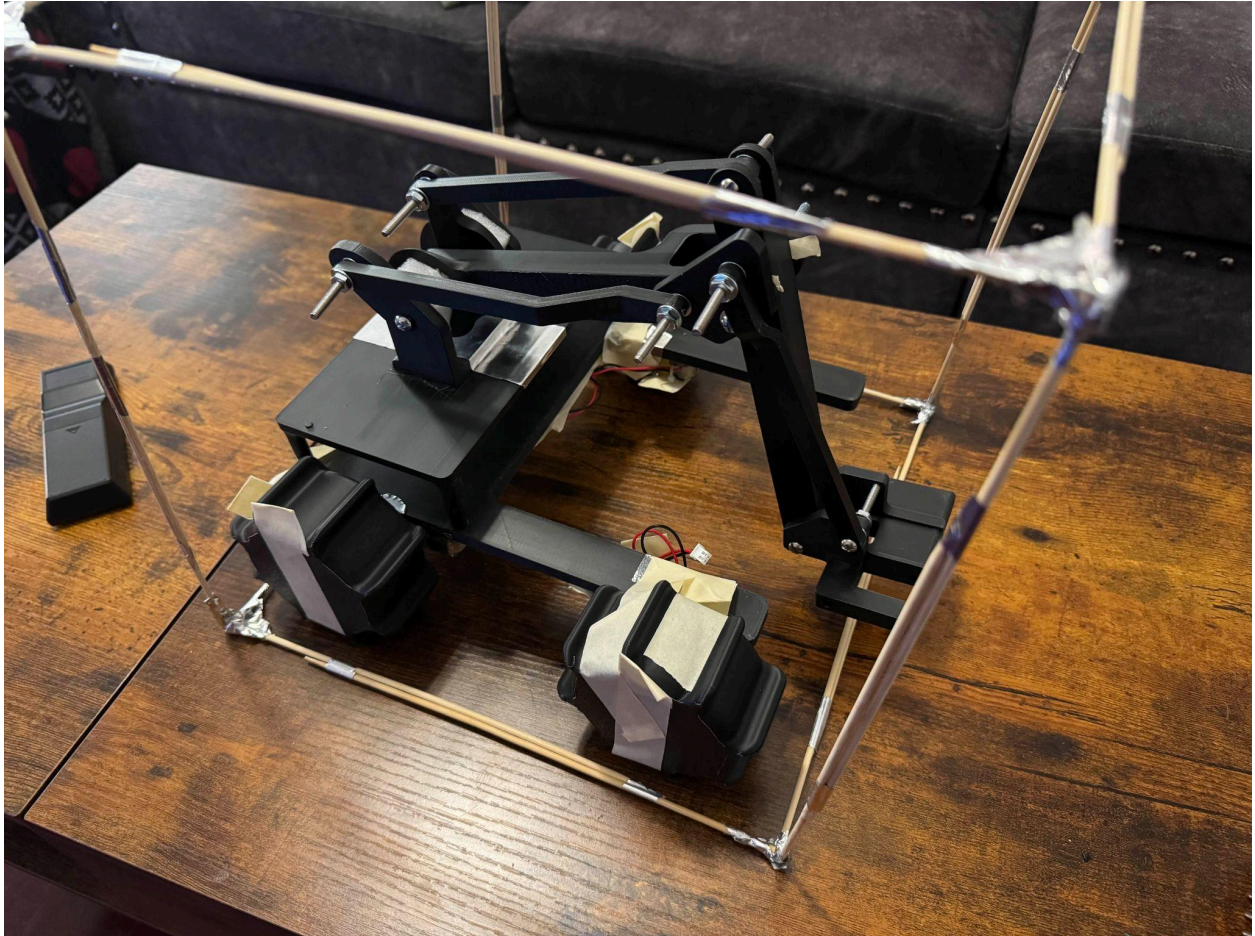


IAM3D FULLY PLA MANUFACTURED R.O.V.E.R “CLAWS”

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ASME
Chapter at Northern Arizona University

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1. Project Objective

The primary objective of this project is the design, manufacture, and testing of a Remote Operations Vehicle for the Excavation of Resources (R.O.V.E.R.). This unmanned vehicle is engineered to navigate a simulated extraterrestrial landscape, identifying and gathering multiple resource types (rocks, metals, and varied materials) and depositing them into a central collection container.

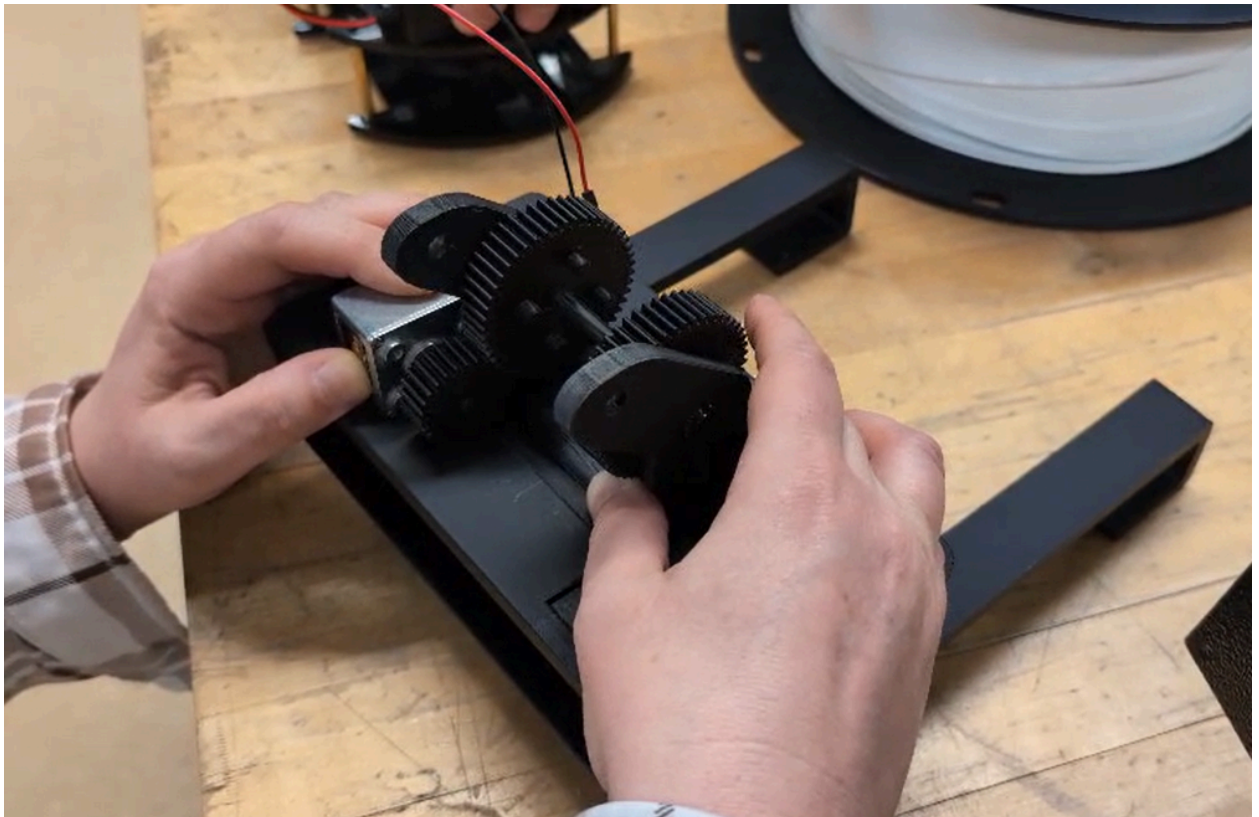
2. Design Process and Iterations

The development of the R.O.V.E.R. followed a rigorous iterative engineering process.

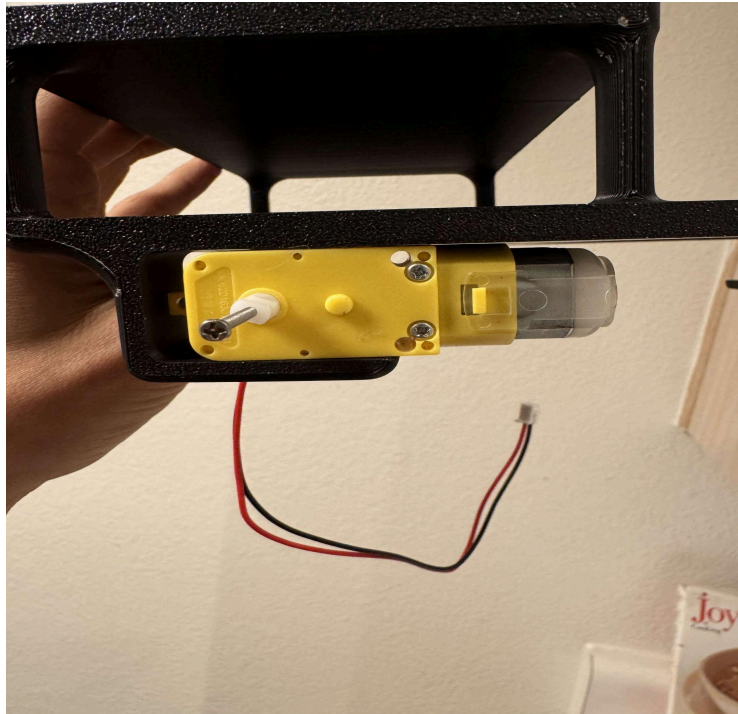
2.1 Initial Prototyping and Testing

The design began with modular testing of key sub-assemblies. Early testing focused on the mechanical advantage of the gear-driven arm and the stability of the chassis.

- **Arm Linkage:** Initial tests of the 3D-printed arm linkages were conducted to ensure proper clearance and range of motion.
- **Gear Mesh:** The gear system for the arm elevation was tested for smooth engagement under load (as seen in photo).

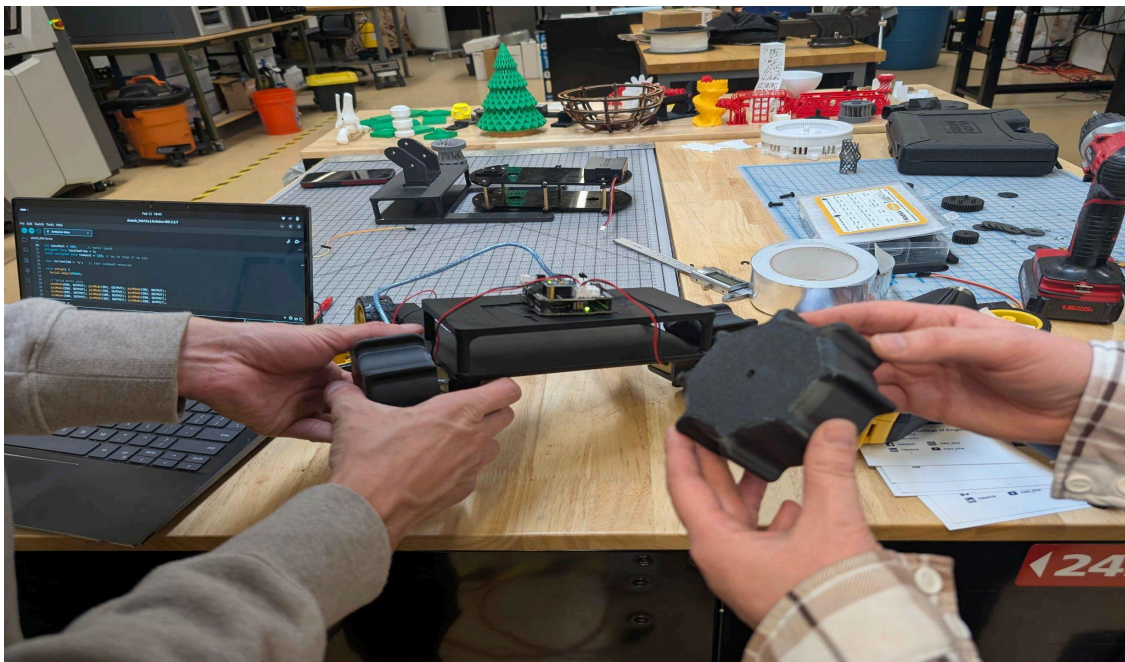


- **Chassis Integration:** The chassis was designed to house the motors internally, providing protection from debris while maintaining a low center of gravity (as seen in photo)



2.2 Design Iterations

- **Iteration 1:** Standard wheel designs were replaced with high-traction, lugged wheels to navigate the sandy course terrain effectively (as seen in photo)



- **Iteration 2:** The excavator bucket was redesigned with a slotted profile to allow fine sand to drain while retaining larger resource pieces.
- **Iteration 3:** Structural reinforcements were added to the arm pivot points to handle the torque required for lifting metal resources.

3. Vehicle Specifications and Constraints

3.1 Size and Compliance

The R.O.V.E.R. is designed to fit within a 30.48cm x 30.48cm x 30.48cm cube during initial inspection. Physical testing with a sizing frame (shown in photo) confirms that the current configuration meets these requirements.



3.2 Additive Manufacturing (AM) Integration

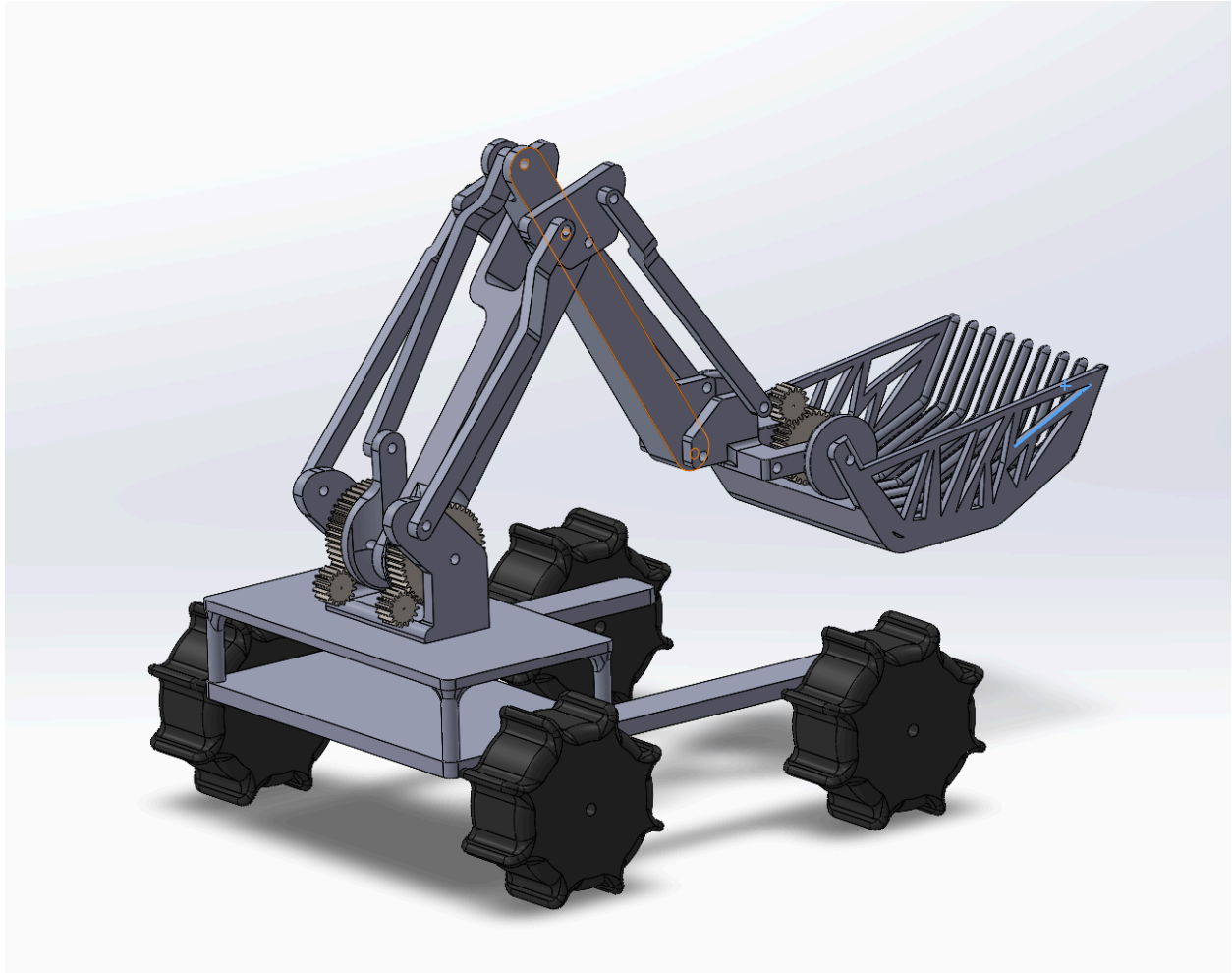
The vehicle maximizes the use of AM parts to optimize the final scoring ratio.

- **Material:** All structural components (chassis, arm links, gears, wheels, and bucket) are manufactured using **Polylactic Acid (PLA)**.
 - **AM Components:** Chassis, four lugged wheels, five arm linkage sections, and one excavator bucket.
 - **Exemptions:** Arduino microcontroller, Raspberry Pi Pico, DC motors, wires, and fasteners.
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4. Systems Architecture

4.1 Mechanical Systems

The R.O.V.E.R. utilizes a four-wheel-drive system powered by individual DC motors geared for high torque. The excavation arm features a multi-jointed design driven by a planetary-style gear set for precise vertical movement and bucket articulation (as shown in the CAD model).

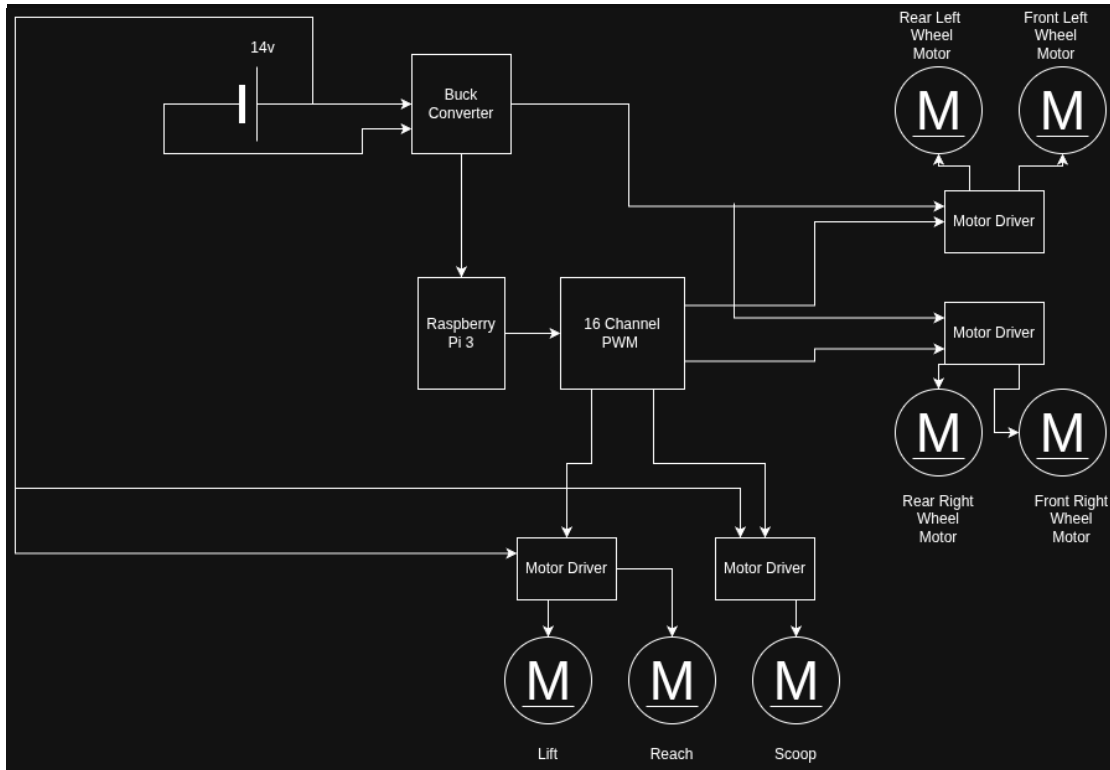


4.2 Electronics and Control

The control system uses a dual-microcontroller architecture as detailed in figure:

- **Command Interface:** A laptop sends inputs to a **Raspberry Pi Pico** over a local server/WiFi connection.
- **Logic Processing:** The Pico processes inputs and instructs the **Arduino** to execute specific motor actions.

- **Power Distribution:** An 8V battery source (complying with the 4S maximum limit) powers the motors through transistors or solid-state relays.



5. Safety and Compliance

The design adheres to all ASME safety provisions:

- **Kill Switch:** A physical kill switch is integrated to cut power to all systems.
- **Wiring:** All powered wires are permanently soldered and insulated with shrink wrap to prevent short circuits.
- **Battery Storage:** LiPo batteries are stored in fire-resistant bags when not in use.
- **Control Safety:** The operator uses a controller with a mandatory arming switch feature.

6. Analyses and Simulations

6.1 Design for Additive Manufacturing (DFAM)

The R.O.V.E.R. was designed with DFAM principles to reduce print time and material waste. Components like the chassis were printed as a single unit to eliminate assembly points, while the bucket was optimized with a lattice structure for a high strength-to-weight ratio.

6.2 Expected Performance

The vehicle is expected to handle resources ranging from 1x1x1 cm to 10x10x10 cm. The high-torque gear ratio in the arm is calculated to lift the heaviest potential metal resources to the required 20.32 cm height for deposition into the central container.